Supporting information for

Possible co-Evolution of Polyglutamine and Polyproline in Huntingtin Protein: Proline Rich Domain as Transient Folding Chaperone

Leili Zhang[1], Hongsuk Kang[1#], Jose Manuel Perez-Aguilar[2], Ruhong Zhou[1,3,4]*

[1]Computational Biology Center, IBM Thomas J. Watson Research Center, Yorktown Heights, NY 10598, USA

[2] School of Chemical Sciences, Meritorious Autonomous University of Puebla (BUAP), University City, Puebla 72570, Mexico

[3]Institute of Quantitative Biology, Zhejiang University, Hangzhou, 310027, China

[3]Department of Chemistry, Columbia University, New York, NY 10027, USA

*All Correspondence should be addressed to: rz24@columbia.edu

[#] Current Address:

University of Maryland – Institute for Bioscience and Biotechnology Research, 9600 Gudelsky Dr., Rockville, MD 20850; National Institute of Science and Technology, 100 Bureau Dr., Gaithersburg, MD 20899

Methods

Molecular dynamics simulation

We obtained the sequences of peptides (DME, DRE, MMU, CFA, BTA, which can be found in Figure 1) based on literature(1). Together with the artificially designed sequences (HQ22-QxA, HQ22-P10A, HQ22Δct, CFA-BTAct, CFAΔct), we prepared the peptides with ArgusLab in linear configuration (ϕ, ψ=180°). Similar to our previous work(2), we used OPLS-AA force field with virtual sites for all hydrogens to enable a larger timestep (4 fs if not specified for all simulations). Note that OPLS-AA force field was chosen based on the force field comparison study on Htt-N17.(3) Periodic boundary conditions were enabled on three directions in all the simulations. For short-range non-bonded interactions, a cutoff distance of 1.2 nm was used for potential shift and Verlet cutoff. Particle mesh Ewald (PME) was used for long-range electrostatic interactions with the Fourier spacing of 0.12 nm. All simulations were performed using GROMACS 5.1.4 software package(4). The linear structures were first collapsed with implicit solvent model GBSA for 1 ns with a timestep of 5 fs. The collapsed structures were then solvated in an 80×80×80 $Å^3$ TIP3P water box and ionized with 150 mM NaCl. After 20000 steps of steepest descent minimization and 1 ns of equilibration with isothermal-isobaric (NpT) ensemble at 310 K, we ran an annealing simulation with canonical ensemble (NVT) under 600 K for 100 ns. Sixteen (16) structures were then selected from this annealing simulation with an interval of 6 ns as the initial structures for the production simulations. The production simulations were conducted with the scheme of Replica Exchange with Solute Tempering (REST(5), enabled by PLUMED(6)) where the van der Waals parameters and partial charges of the solute (peptides in this case) were scaled in each replica. Sixteen (16) replicas were used with the scaling factor of (1.0, 0.967, 0.933, 0.9, 0.867, 0.833, 0.8, 0.767, 0.733, 0.7, 0.667, 0.633, 0.6, 0.567, 0.533, 0.5). The average exchange rate between replicas was determined to be 32 ±2%. The production simulations were performed for 400 ns for each replica. Structures were recorded every 0.1 ns for the data analyses, for which the first 100 ns of simulations were discarded. We then used the RMSD clustering method provided by VMD(7) to estimate the population of sampled structures for each of the peptides. The cluster centers were illustrated using their secondary structures in Figure 2, 3, and 4 in the format of "tree maps" (where cluster sizes are proportional to the block sizes in the map). In Figure S15, we plotted free energy landscape of all peptides simulated in this study using secondary structure contents (alpha score and beta score, see below) as the order parameters. We assessed the convergence of our simulations by comparing the first half of the simulations and the second half of the simulations with the entire trajectory. As we can see, the free energy basins were shallow (~2 kcal/mol) and remained generally stable between the first half and the second half of the simulations.

Shape anisotropy

We use the following equation to calculate shape anisotropy (8) κ for the polyQ domains:

$$\kappa^2 \equiv \frac{3}{2}\frac{\lambda_x^4+\lambda_y^4+\lambda_z^4}{\left(\lambda_x^2+\lambda_y^2+\lambda_z^2\right)^2}-\frac{1}{2} \tag{1}$$

where $\lambda_m = \sum_i M_i r_{i,m}^2 / \sum_i M_i$ when axes m are the principle axes of the protein, M_i is the mass of atom i and r_i is the position of atom i. When the protein is perfectly linear, κ=1; and when it is perfectly spherical, κ=0.

Alpha/beta score

The alpha/beta score is a quantitative measurement of the similarity between a certain protein structure and a perfect alpha helix or a perfect beta sheet. We directly used Pietrucci et al.'s implementation(9) in PLUMED to calculate this score. In summary, the alpha score is directly from ALPHARMSD, while the beta score is the summation of anti-beta score (ANTIBETARMSD) and para-beta score (PARABETARMSD) in PLUMED. Each individual score (s) was calculated using:

$$s = \sum_i \frac{1 - \left(\frac{r_i - d_0}{r_0}\right)^n}{1 - \left(\frac{r_i - d_0}{r_0}\right)^m}$$

where r_i is the coordinate of ith atom from the structure of interest, d_0 is the coordinate of ith atom from the reference structure (alpha helix, parallel beta sheet or antiparallel beta sheet), r_0 is a parameter (0.08 nm), m is a parameter set to 12 and n is a parameter set to 8. The calculations are iterated through all possible sets of six consecutive sequences for alpha scores, or two consecutive sequences each consisting three residues from the protein structure for beta scores. For example, we calculated the NQ beta score by adding para-beta score and anti-beta score which was computed using all possible three plus three consecutive residues in the N17 and polyQ region (MAT+QQQ, etc.). Note that because of the definition, the NQ beta score is not a simple addition of the N17 beta score and polyQ beta score. But instead, it also includes the cross term between N17 and polyQ.

PPII helical content in polyQ seems to be unaffected by the deletion of PRD

We tested a competing theory presented in literature(10-12), which proposed that polyP in PRD provided a protective role by inducing the structures of polyQ to form PPII helical structures through backbone torsional propagation. To examine the impacts of PRD, we compared the PPII helical content in a fixed length of polyQ region between proteins that contained PRD (HQ22 and CFA) and proteins that did not contain PRD (HQ22Δct and CFAΔct). Only the 5 glutamine residues on the C-terminus (C5Q) were examined, because they were directly connected to PRD through the backbone. We took the definition of PPII helix from Chebrek et al.(13), which defined a PPII helix as two consecutive residues adopting phi angle values from -104° to -46° and psi angle values from 116° to 174°. We found that the average PPII helical residue count of C5Q was 0.137 for HQ22 and 0.027 for HQ22Δct, while the average PPII helical residue count of C5Q was 0.033 for CFA and 0.058 for CFAΔct (block average standard deviations are around 0.010). The small number of PPII helical content in C5Q indicated that the PPII helix in these residues was transient in nature. A significant decrease of the PPII helical content in HQ22-C5Q was detected when PRD was deleted. However, the CFA-C5Q PPII helical content was slightly increased when PRD was deleted. As a result of this contradiction, within the scope of our current study, we concluded that backbone inducement of PPII helix in polyQ was not observed. Our calculated values of low PPII helical content did not necessarily disagree with the Circular Dichroism spectra data in the aforementioned studies (10-12), because the PPII helix in some peptides could be transient but dominating in the CD spectra (14, 15).

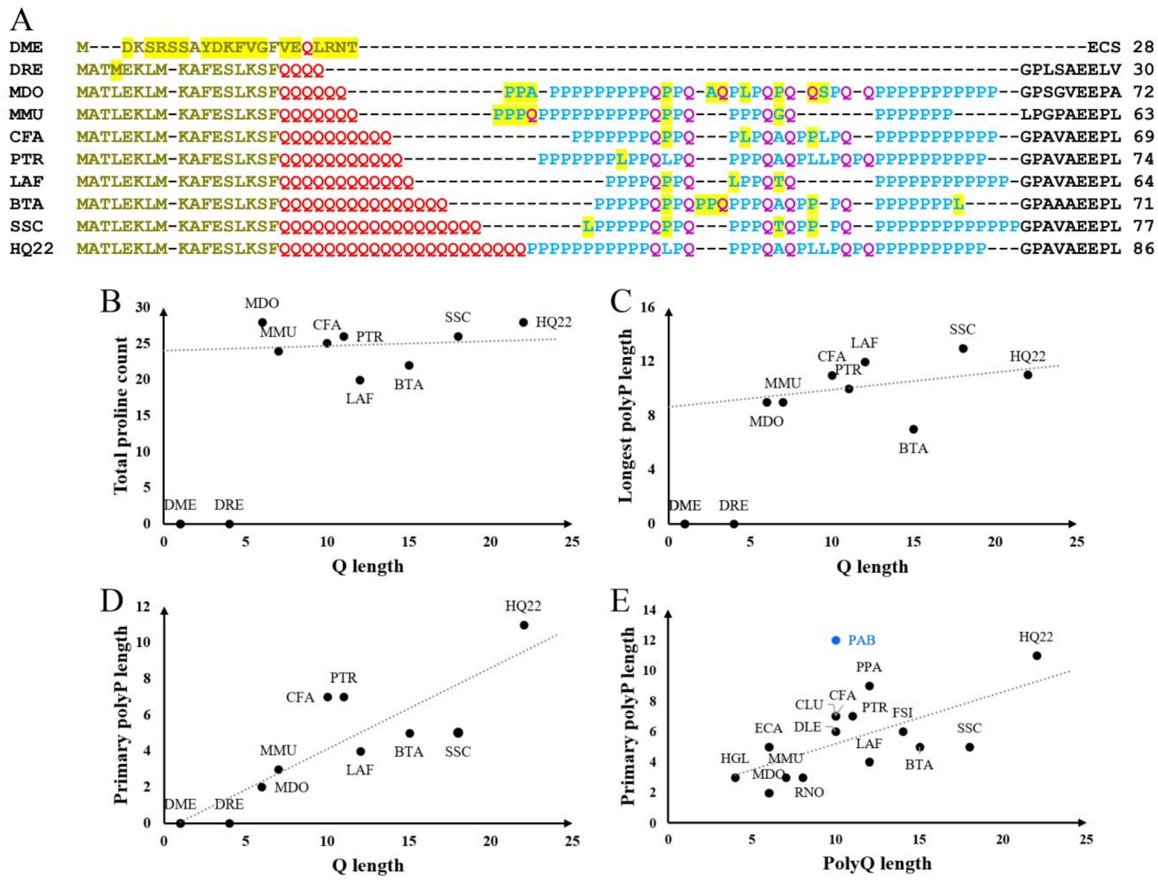

Figure S1. An expanded sequence alignment of the N-terminus (hypothetical exon-1) of Htt proteins are shown in (A). From top to bottom: *Drosophila melanogaster* (DME), *Danio rerio* (DRE), *Monodelphis domestica* (MDO), *Mus musculus* (MMU), *Canis familiaris* (CFA), *Pan troglodytes* (PTR), *Loxodonta africana* (LAF), *Bos taurus* (BTA), *Sus scrofa* (SSC) and *Homo sapiens* (HQ22, with 22 Qs). N17s are colored in tan, polyQs are colored in red, PRDs (except PRD-Qs) are colored in cyan, PRD-Qs are colored in purple and sequences after PRDs are colored in black. To clearly compare the number of Qs in polyQ and number of Ps in PRD, we plotted the total P count (B), the longest polyP length (C) and the primary polyP length (D, defined as the polyP stretch that is closest to the polyQ in sequence) against the polyQ length. The dashed lines in (B-D) are guide to the eyes, which are calculated using linear regression on all species except DME and DRE. Correlation coefficient R^2 was calculated to be 0.02, 0.04 and 0.76 for (B), (C) and (D), respectively. In subplot (E), we plotted primary polyP length against polyQ length with a further expanded list of species that were examined in a recent study by Seefelder et al.(16) We have added the following species to the plot: ECA (*Equus caballus*), PPA (*Panthera pardus*), FSI (*Felis silvestris catus*), DLE (*Delphinapterus leucas*), CLU (*Canis lupus familiaris*), HGL (*Hetercephalus glaber*), and PAB (*Pongo abelii*). To show the correlation among mammalian vertebrates, non-mammalian vertebrates and non-vertebrates (species that were found to have 4 or less Qs and no PRD, such as DME and DRE) were removed from the plot. The dashed line was calculated using linear regression among all species ($R^2 \sim 0.45$). PAB was excluded as an outlier for the following reasons. Their primary polyP was long, perhaps because they were close relatives to PTR and human. However, the particular entry (XP_024101641.1) might possess particularly short polyQ due to polymorphism. Needless to say, more sequencing data are needed to elucidate on such correlations.

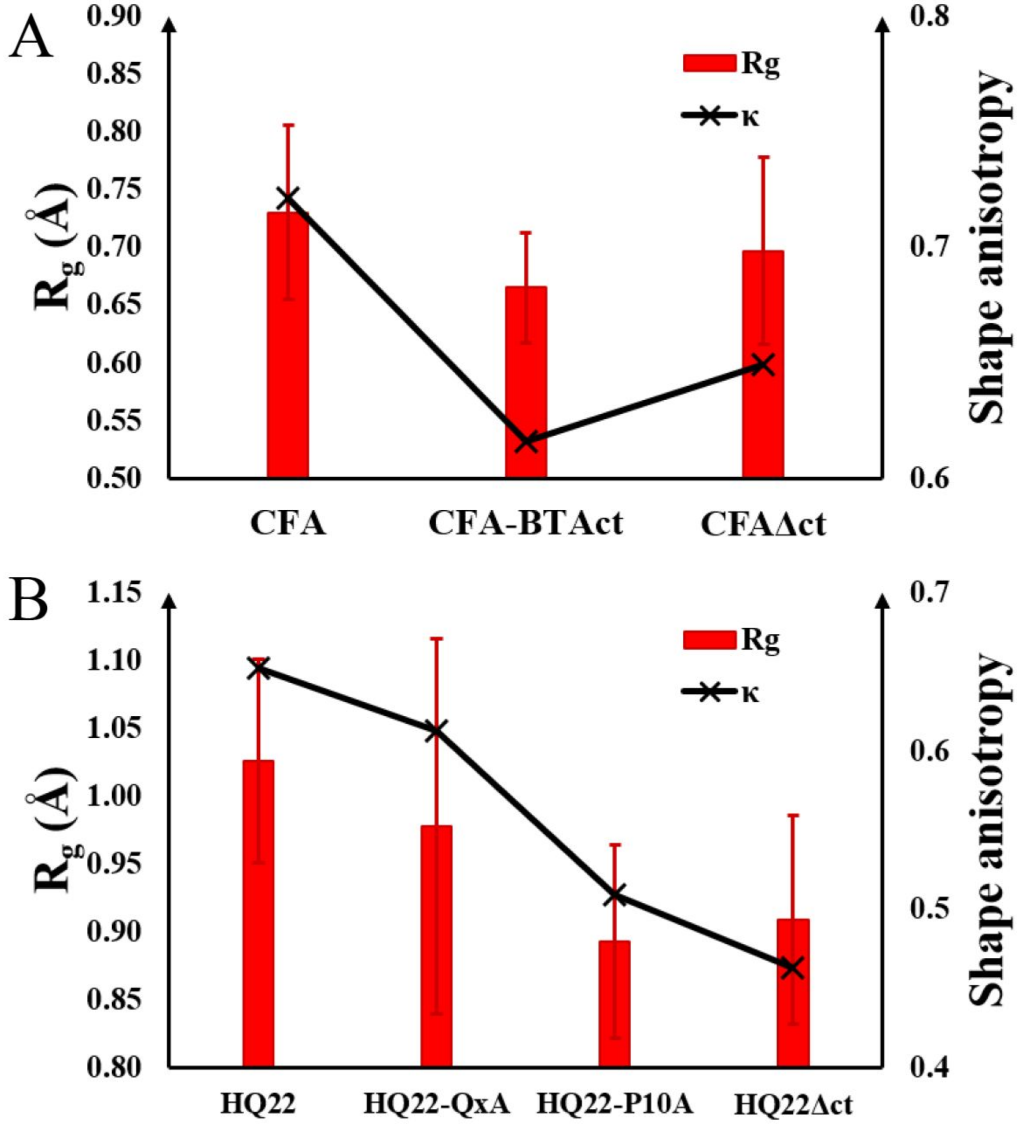

Figure S2. Radius of gyration and relative shape anisotropy for the CFA peptides (CFA, CFA-BTAct and CFAΔct, in A) and HQ22 peptides (HQ22, HQ22-QxA, HQ22-P10A and HQ22Δct, in B).

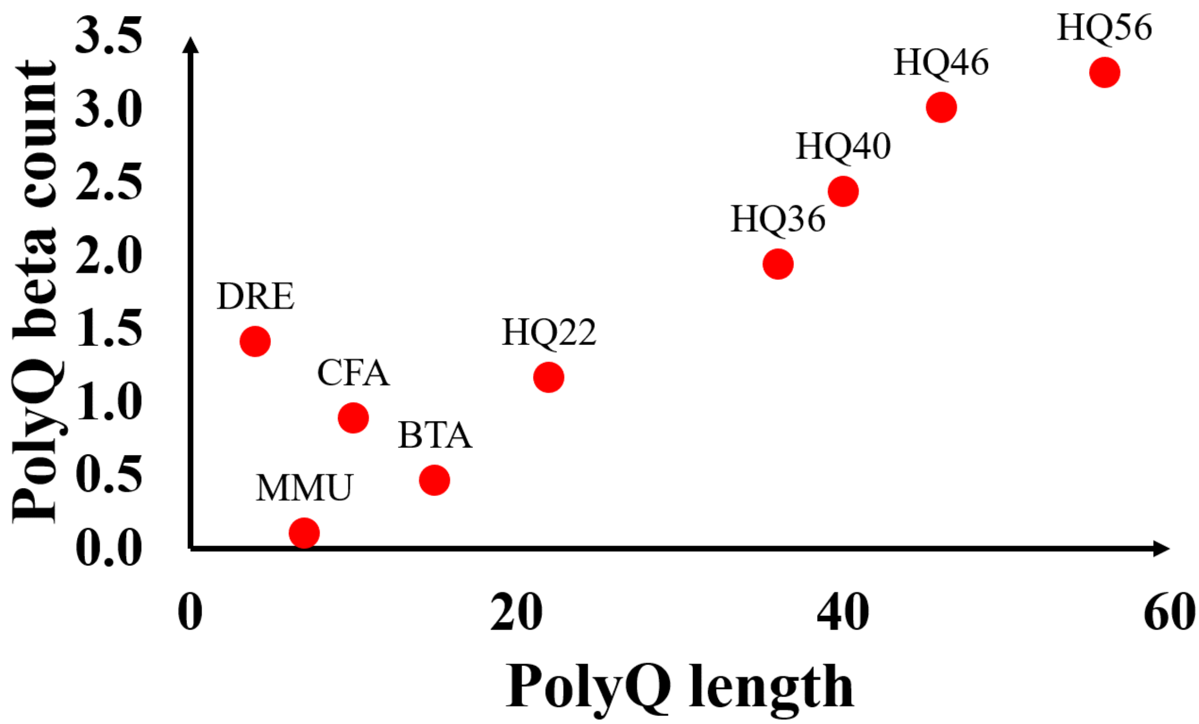

Figure S3. The average number of residues in beta structures is plotted against the Q length in polyQ domains.

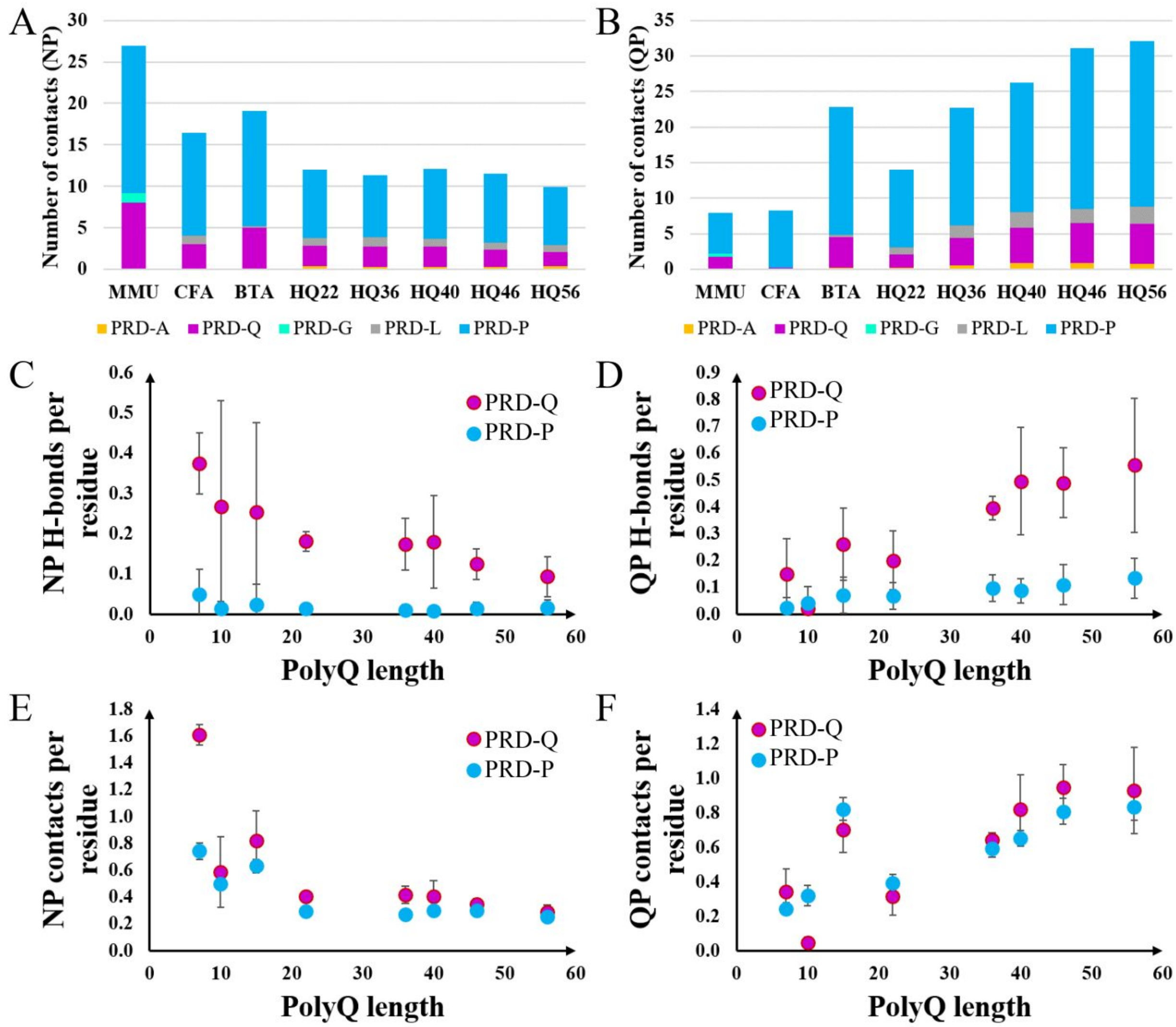

Figure S4. Average numbers of NP heavy atom contacts (within 4.5Å) per timeframe are plotted in (A). Average numbers of QP heavy atom contacts (within 4.5Å) per timeframe are plotted in (B). Contributions made from various residues in PRD are colored in: yellow (alanines, i.e. PRD-As), purple (glutamines, i.e. PRD-Qs), lime (glycines, i.e. PRD-Gs), gray (leucines, i.e. PRD-Ls), cyan (prolines, i.e. PRD-Ps). Normalized H-bonds stemmed from PRD-Q (purple) and PRD-P (cyan) per timeframe for NP (C, which are between N17 and PRD) and QP (D, which are between polyQ and PRD). Normalized residue contacts within 4.5 Å stemmed from PRD-Q (purple) and PRD-P (cyan) per timeframe for NP (E) and QP (F). All numbers are normalized with the number of Qs or Ps in the PRD of the corresponding peptide, so the result reflects H-bonds/contacts per residue for PRD-Qs or PRD-Ps.

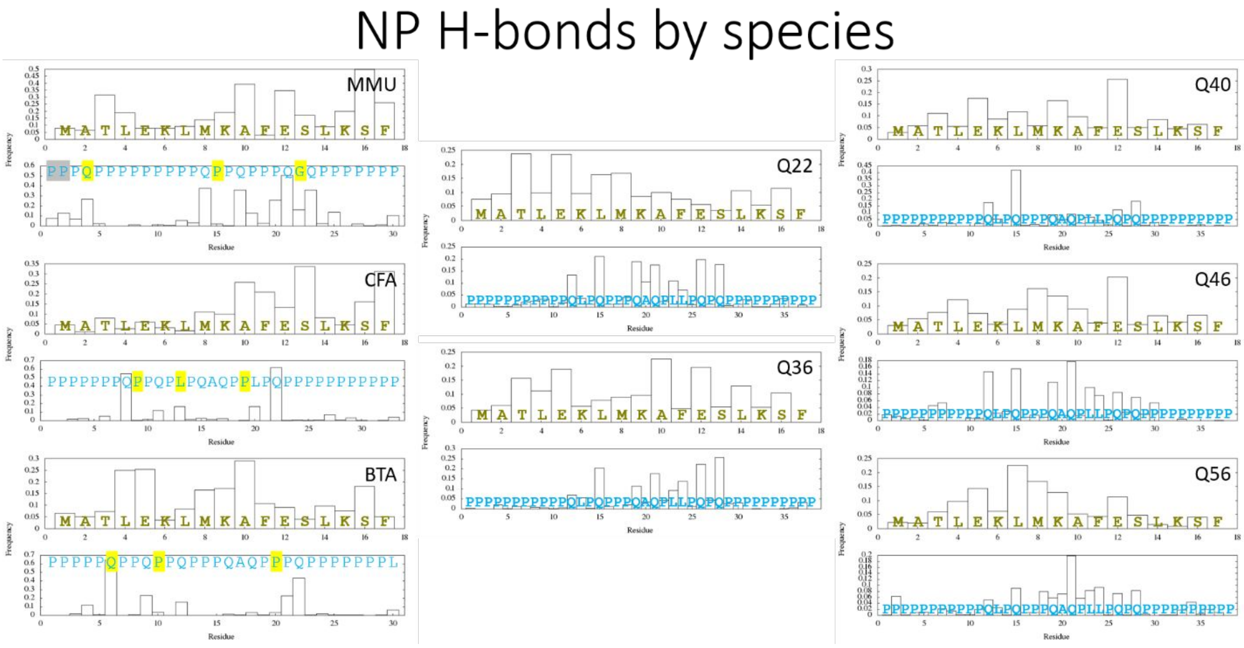

Figure S5. We summarize the frequency of a particular residue involved in NP H-bonds for all wildtype simulations. For example, M1 of MMU-N17 is involved in any NP H-bonds during 8% of the simulation of MMU HttEx1. Hydrophilic residues of N17 and PRD-Qs are clearly more frequently involved in NP H-bonds.

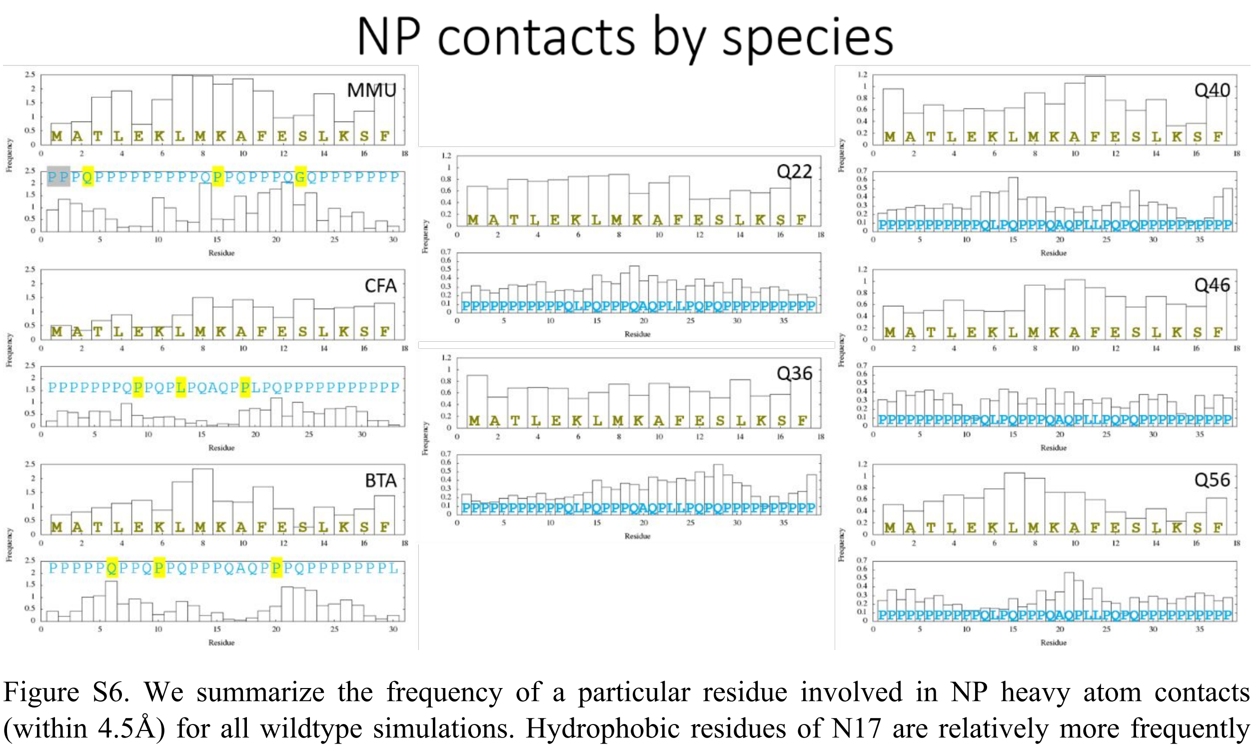

Figure S6. We summarize the frequency of a particular residue involved in NP heavy atom contacts (within 4.5Å) for all wildtype simulations. Hydrophobic residues of N17 are relatively more frequently involved in NP contacts.

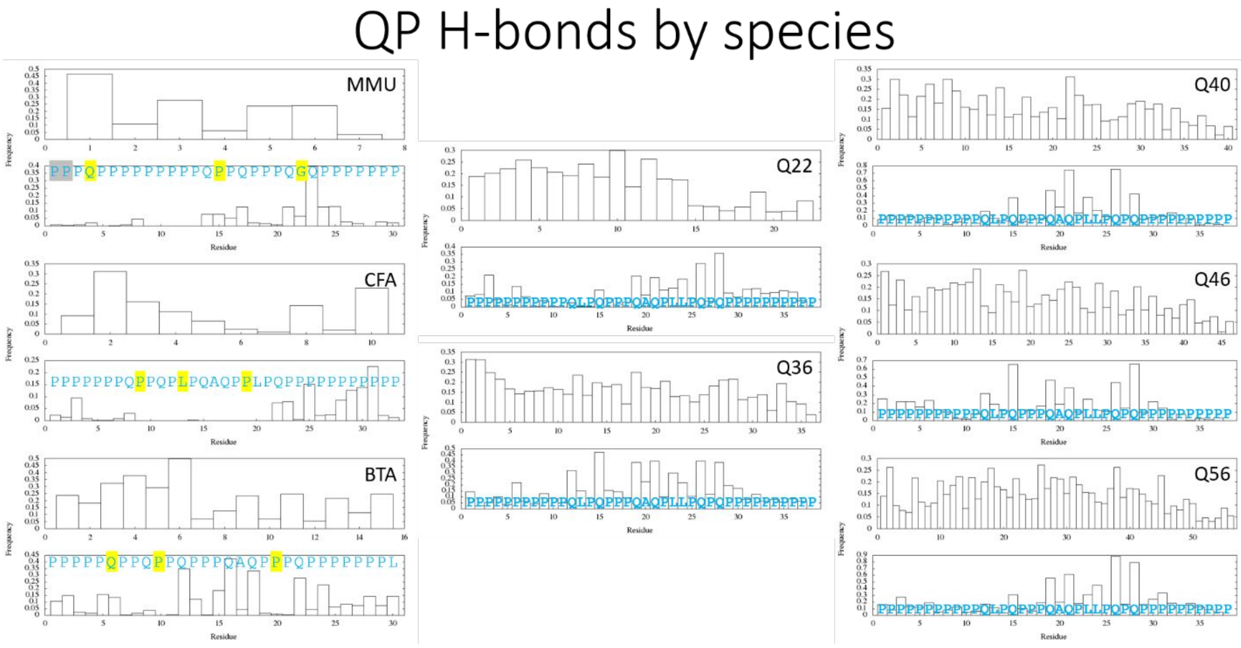

Figure S7. We summarize the frequency of a particular residue involved in QP H-bonds for all wildtype simulations. PRD-Qs are clearly more frequently involved in QP H-bonds.

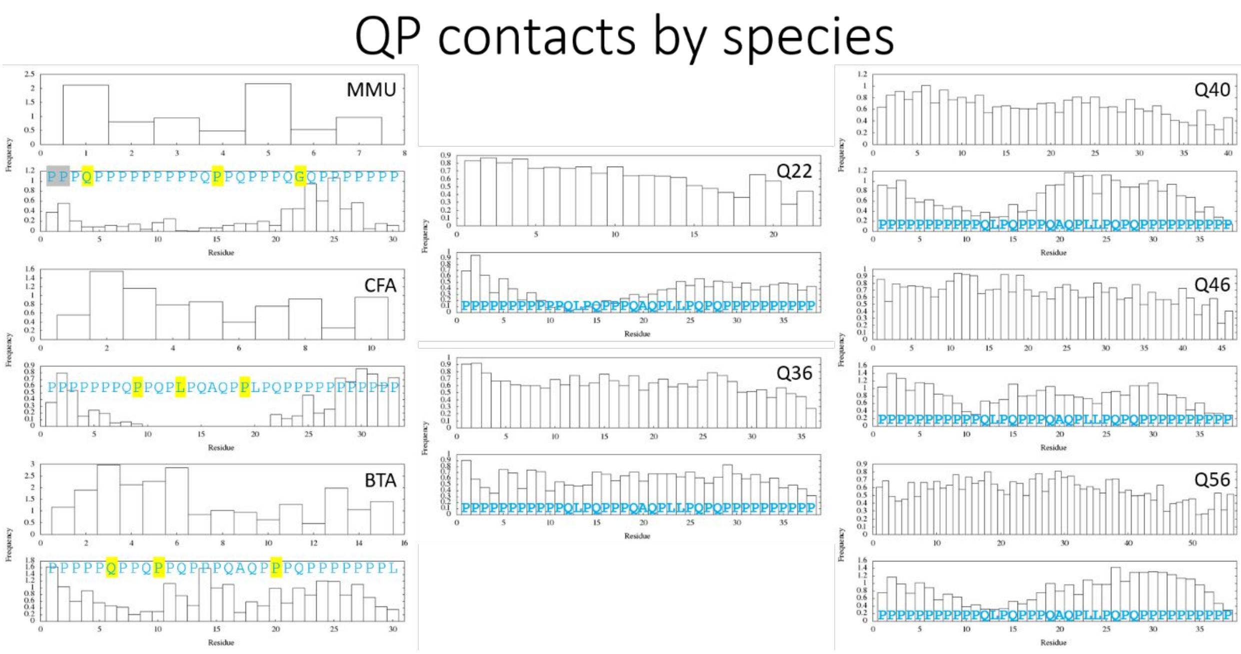

Figure S8. We summarize the frequency of a particular residue involved in QP heavy atom contacts (within 4.5Å) for all wildtype simulations. Longer stretches of polyP (including the primary polyP) are relatively more frequently involved in QP contacts.

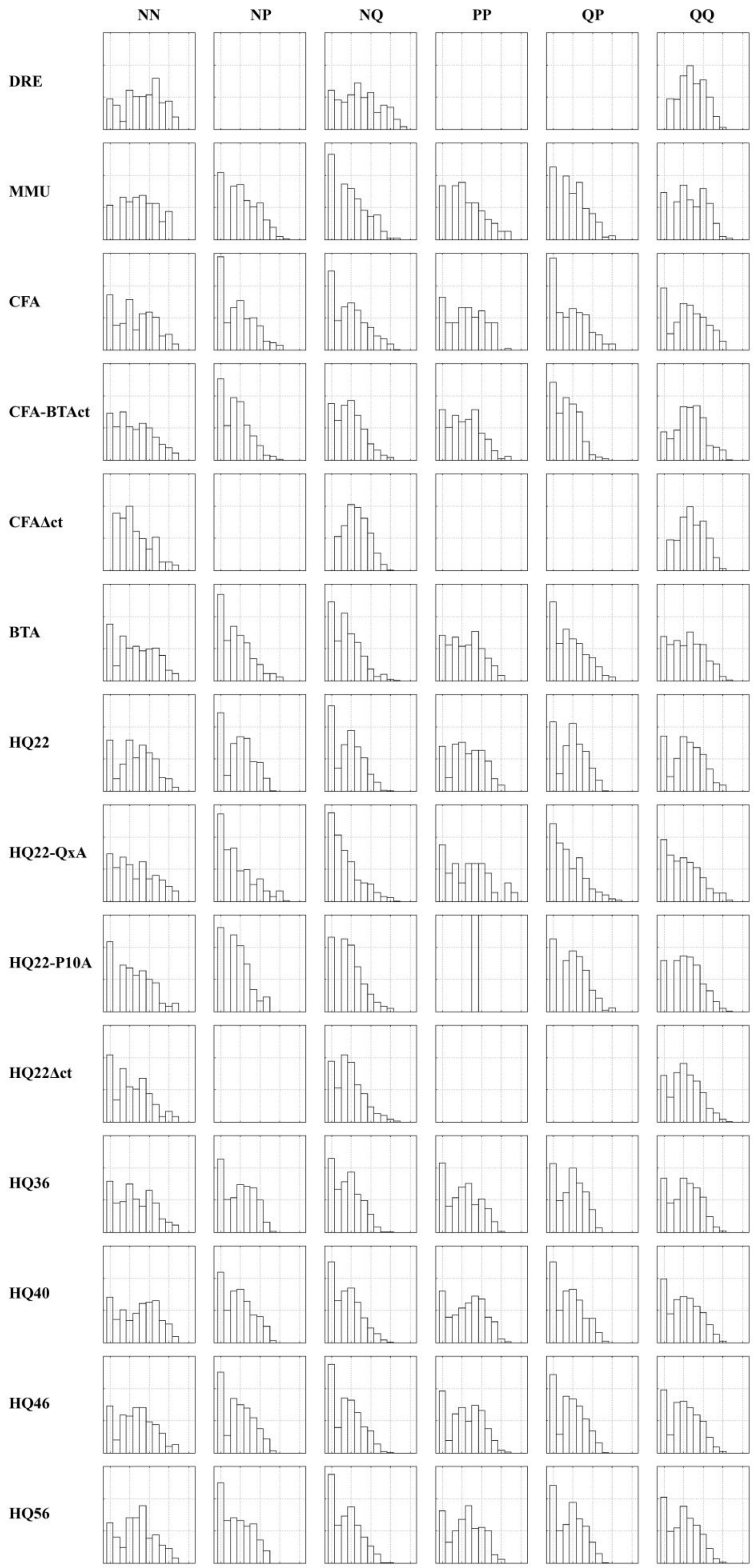
NN
NP
NQ
PP
QP
QQ
DRE
MMU
CFA
CFA-BTAct
CFAΔct
BTA
HQ22
HQ22-QxA
HQ22-P10A
HQ22Δct
HQ36
HQ40
HQ46
HQ56

Figure S9. We summarize the number of interdomain H-bonds (y axes) with different persistence time (x axes) in this plot of histograms. For each peptide, interdomain H-bonds are splat into six types: N17-N17 (NN), N17-PRD (NP), N17-polyQ (NQ), PRD-PRD (PP), polyQ-PRD (QP) and polyQ-polyQ (QQ). For each subplot, the y axes are normalized by the total number of H-bond pairs for each interdomain interactions and each species. Each tick is 10%, summing up to 30% in maximum for all subplots. The x axes are the logarithms of the persistence time, with ticks corresponding to 0.1 ns, 1 ns, 10 ns, 100 ns and 1000 ns from left to right.

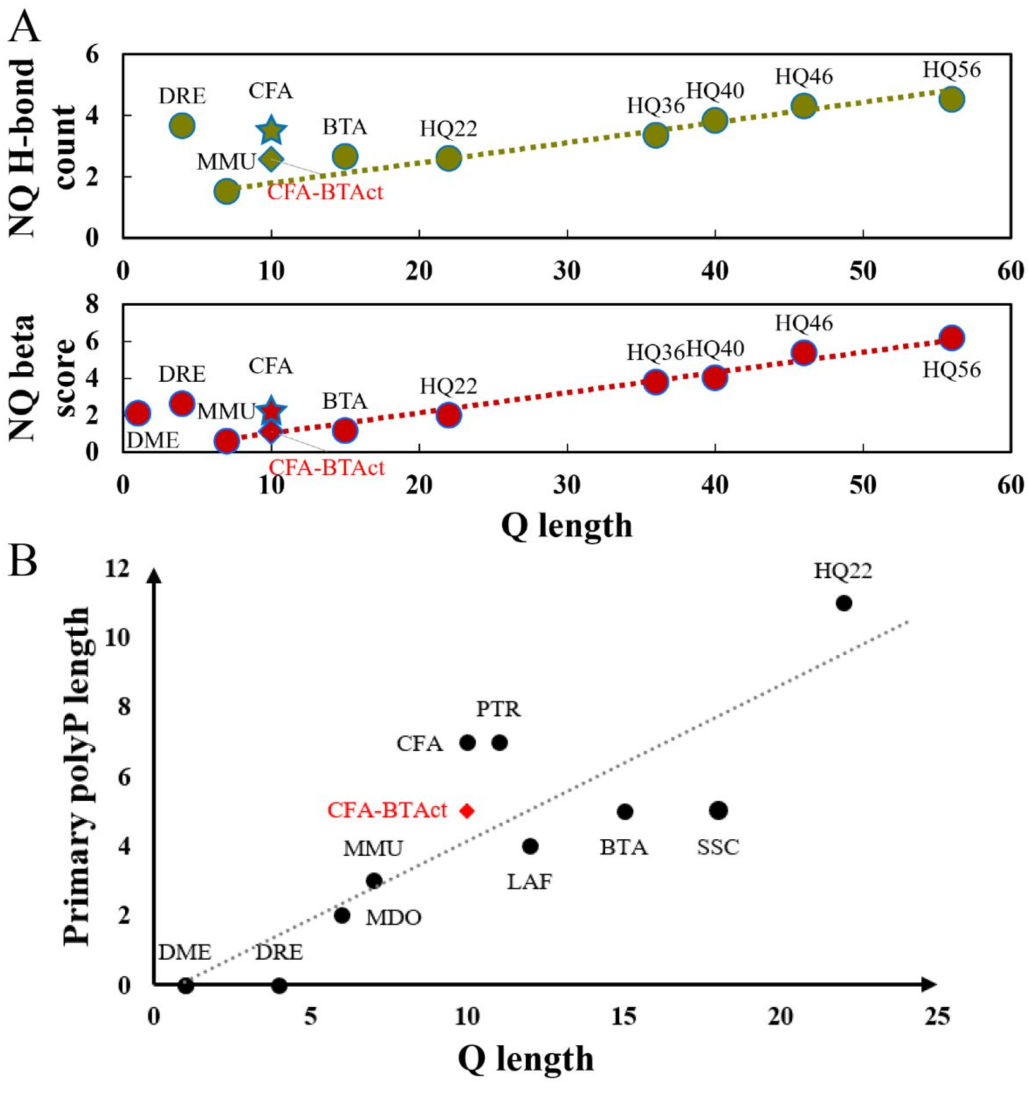

Figure S10. (A) Numbers of NQ H-bonds are plotted for various species with the addition of CFA-BTAct. NQ beta scores are plotted for various species with the addition of CFA-BTAct. (B) The primary polyP length is plotted against the Q length for various species, with the addition of CFA-BTAct.

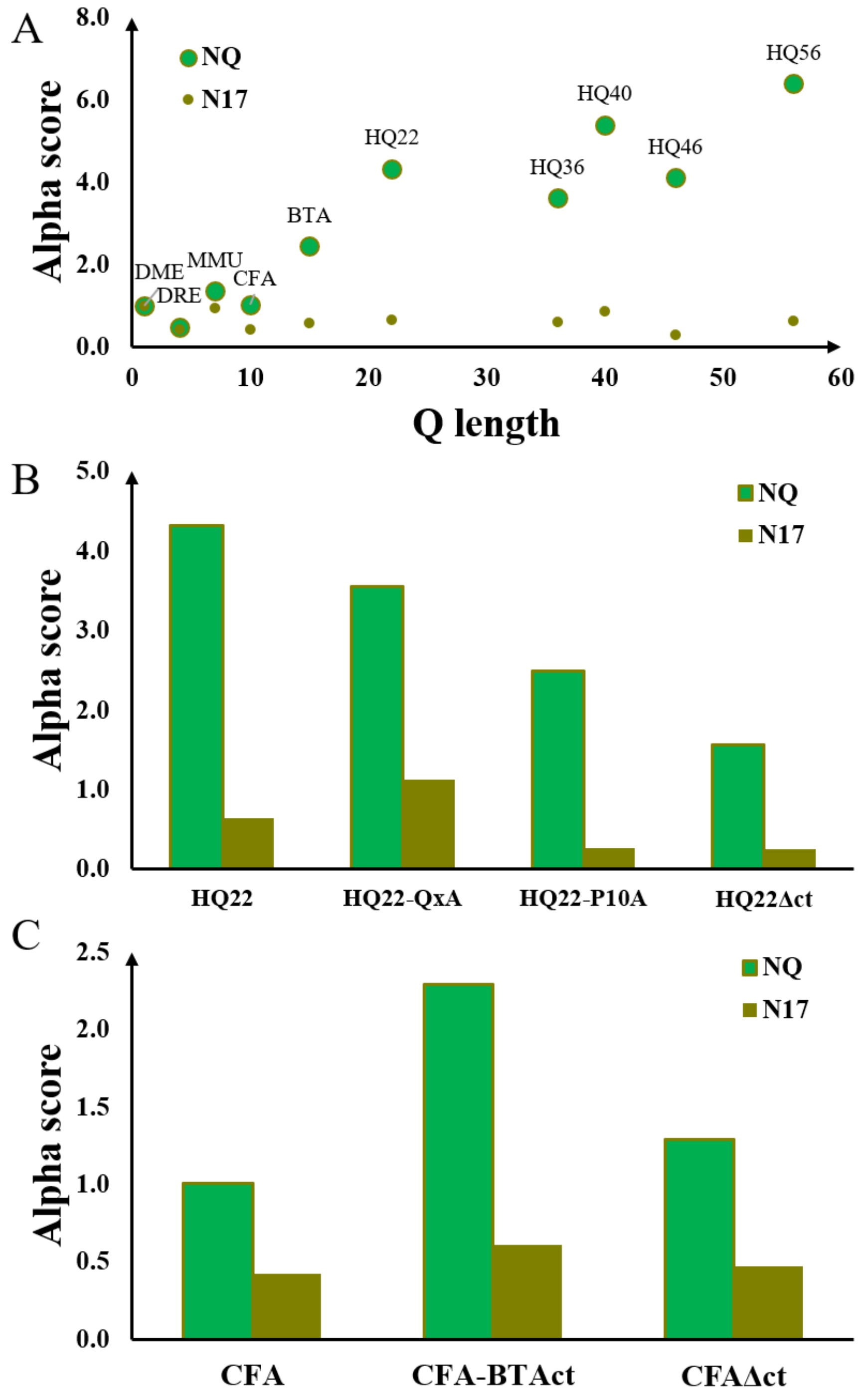

Figure S11. Alpha scores of N17-polyQ (NQ) and N17 only are plotted for all wildtype simulations (DME, DRE, MMU, CFA, BTA, HQ22, HQ36, HQ40, HQ46, HQ56) in A, HQ22 peptides (HQ22, HQ22-QxA, HQ22-P10A and HQ22Δct) in B, CFA peptides (CFA, CFA-BTAct and CFAΔct) in C.

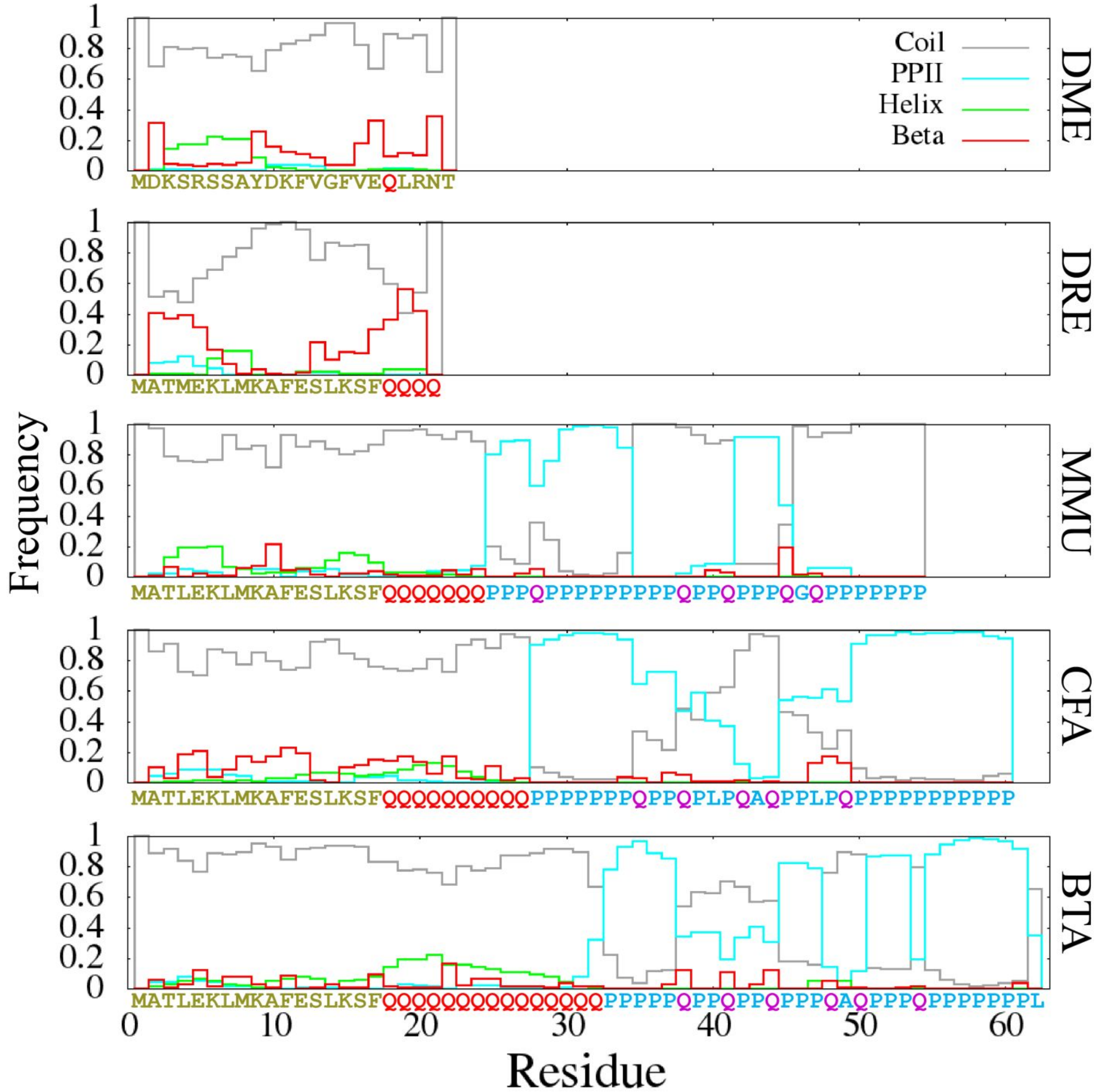

Figure S12. The average percentage of the secondary structure components is plotted for all residues of DME, DRE, MMU, CFA and BTA. The plotted lines are colored in gray for random coils, cyan for polyproline II (PPII) helices, red for helical structures (alpha helices or 3-10 helices) and green for beta structures (beta sheets or beta bridges).

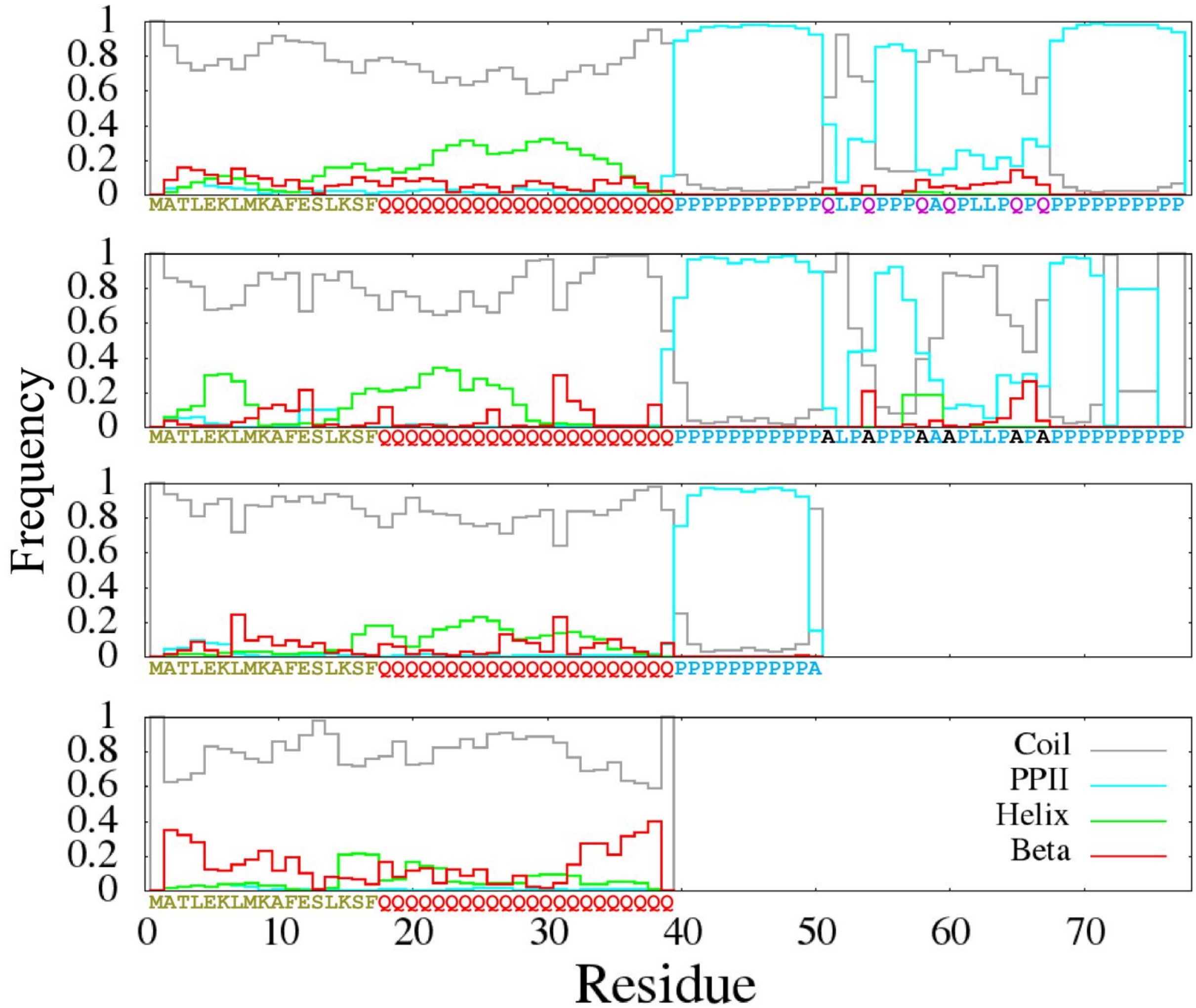

Figure S13. The average percentage of the secondary structure components is plotted for all residues of HQ22, HQ22-QxA, HQ22-P10A and HQ22Δct (from top to bottom). The plotted lines are colored in gray for random coils, cyan for polyproline II (PPII) helices, red for helical structures (alpha helices or 3-10 helices) and green for beta structures (beta sheets or beta bridges).

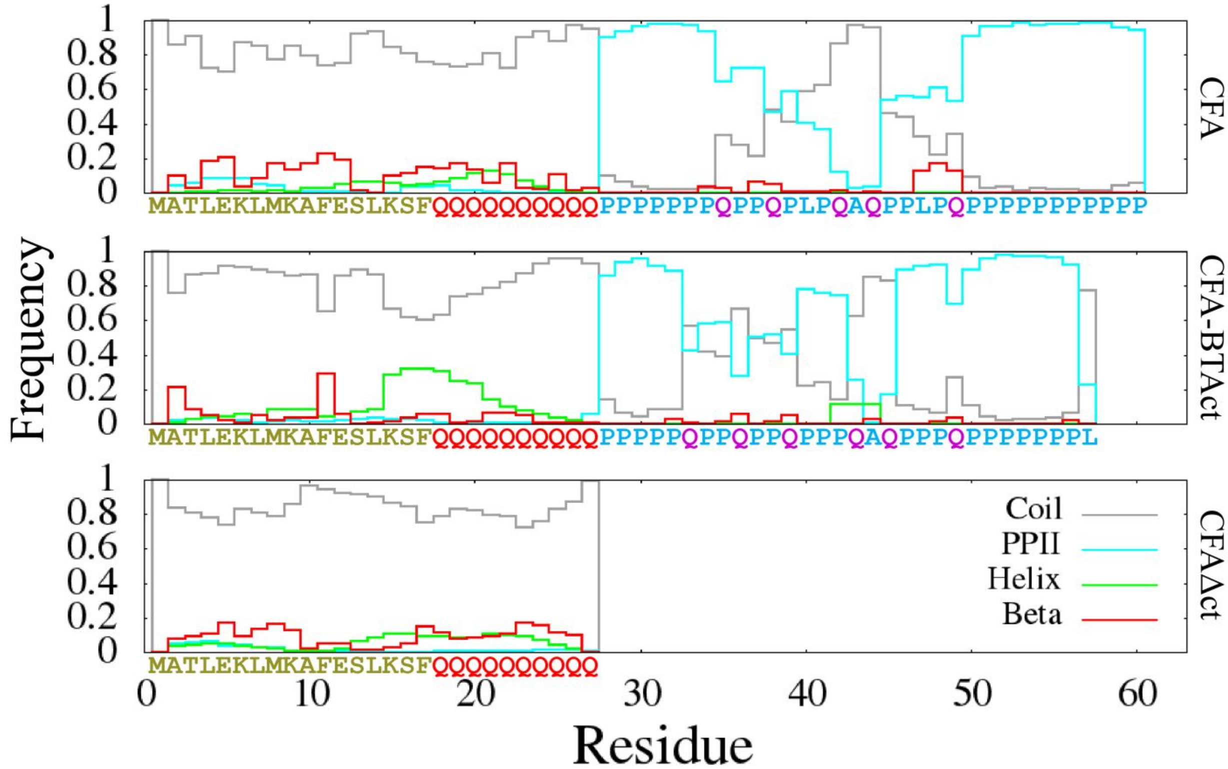

Figure S14. The average percentage of the secondary structure components is plotted for all residues of CFA, CFA-BTAct and CFAΔct (from top to bottom). The plotted lines are colored in gray for random coils, cyan for polyproline II (PPII) helices, red for helical structures (alpha helices or 3-10 helices) and green for beta structures (beta sheets or beta bridges).

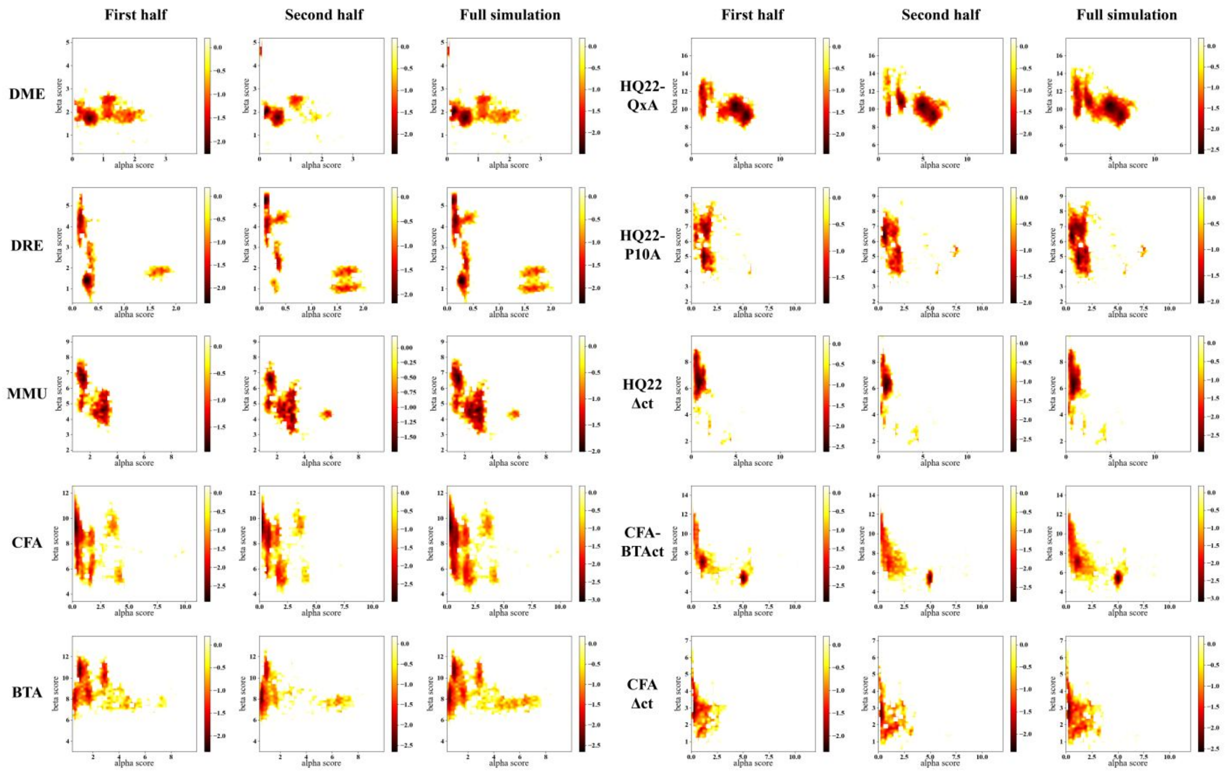

Figure S15. Free energy landscape calculated with a Boltzmann function using the alpha score (x-axes) and the beta score (y-axes) of the HttEx1 proteins examined in this study. To show the convergence of our ~400 ns simulations, we plotted the free energy landscape of the first half, second half, and the entire trajectory of the simulations in three columns after deleting the initial 100 ns of equilibrations.

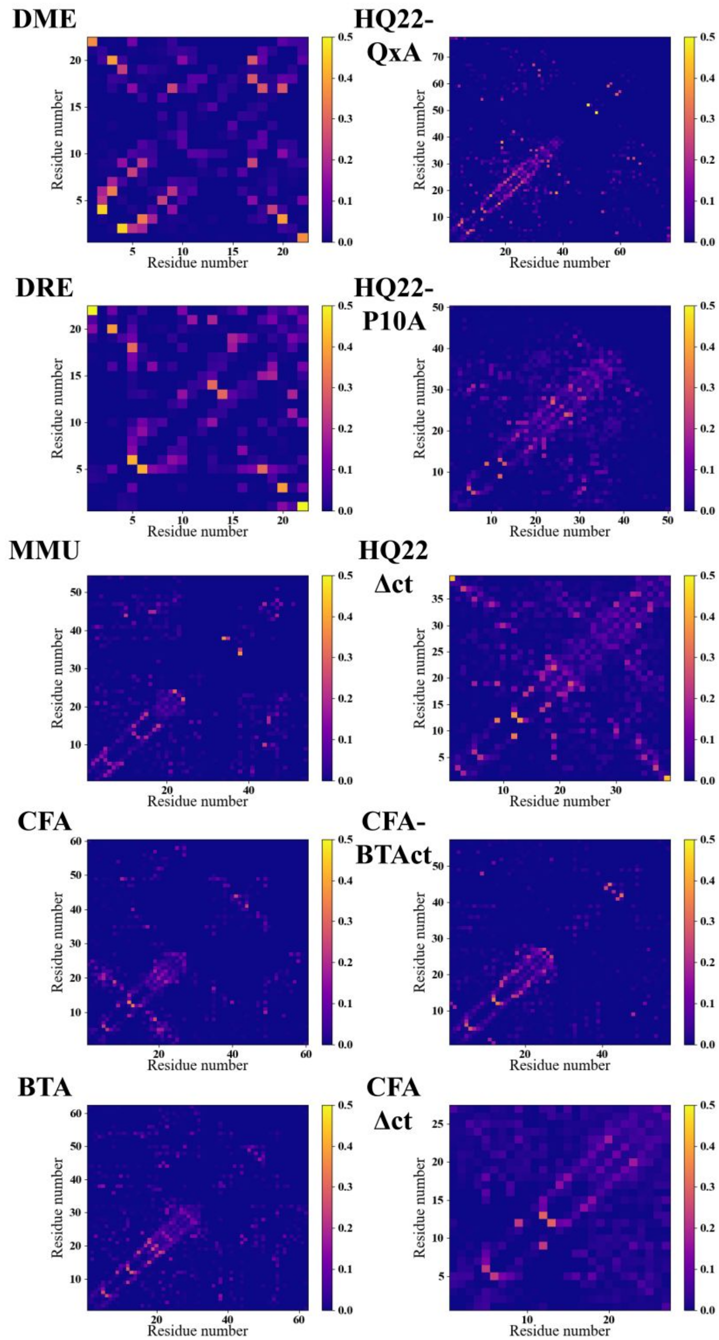

Figure S16. Hydrogen bond frequency map between any residue pairs in the monomeric HttEx1 proteins simulated in this study. The frequency values were calculated as $\Sigma_T \delta(Hbond_{M-N})/T$, where T denotes the total number of frames of the simulation, delta function $\delta(Hbond_{M-N})$ is 1 if any hydrogen bond between residue M and N exists and 0 if no hydrogen bond between residue M and N exists.

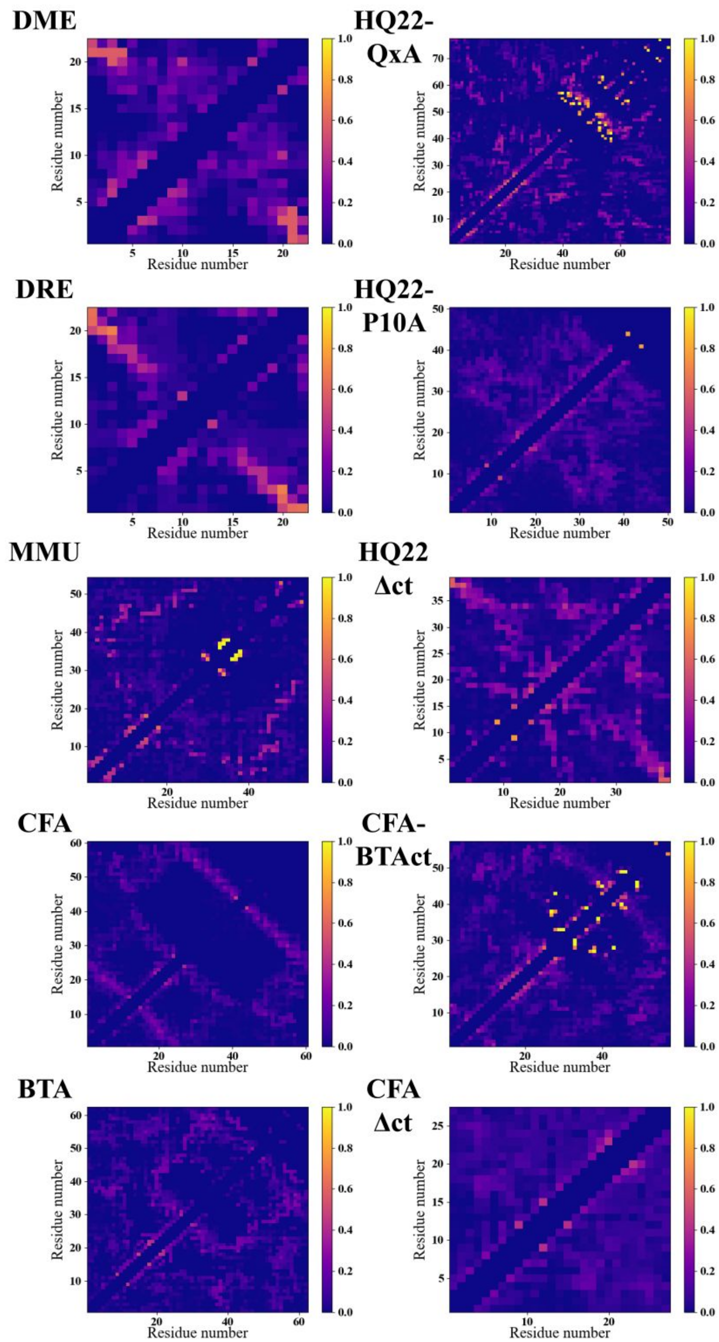

Figure S17. Contact frequency map between any residue pairs in the monomeric HttEx1 proteins simulated in this study. The frequency values were calculated as $\Sigma_T \delta(Contact_{M-N})/T$, where T denotes the total number of frames of the simulation, delta function $\delta(Contact_{M-N})$ is 1 if any heavy atom distances between residue M and N were below 4.5 Å and 0 if no heavy atom distances between residue M and N were below 4.5 Å. Contacts between residues that were within 3 residues in sequence to each other were excluded.

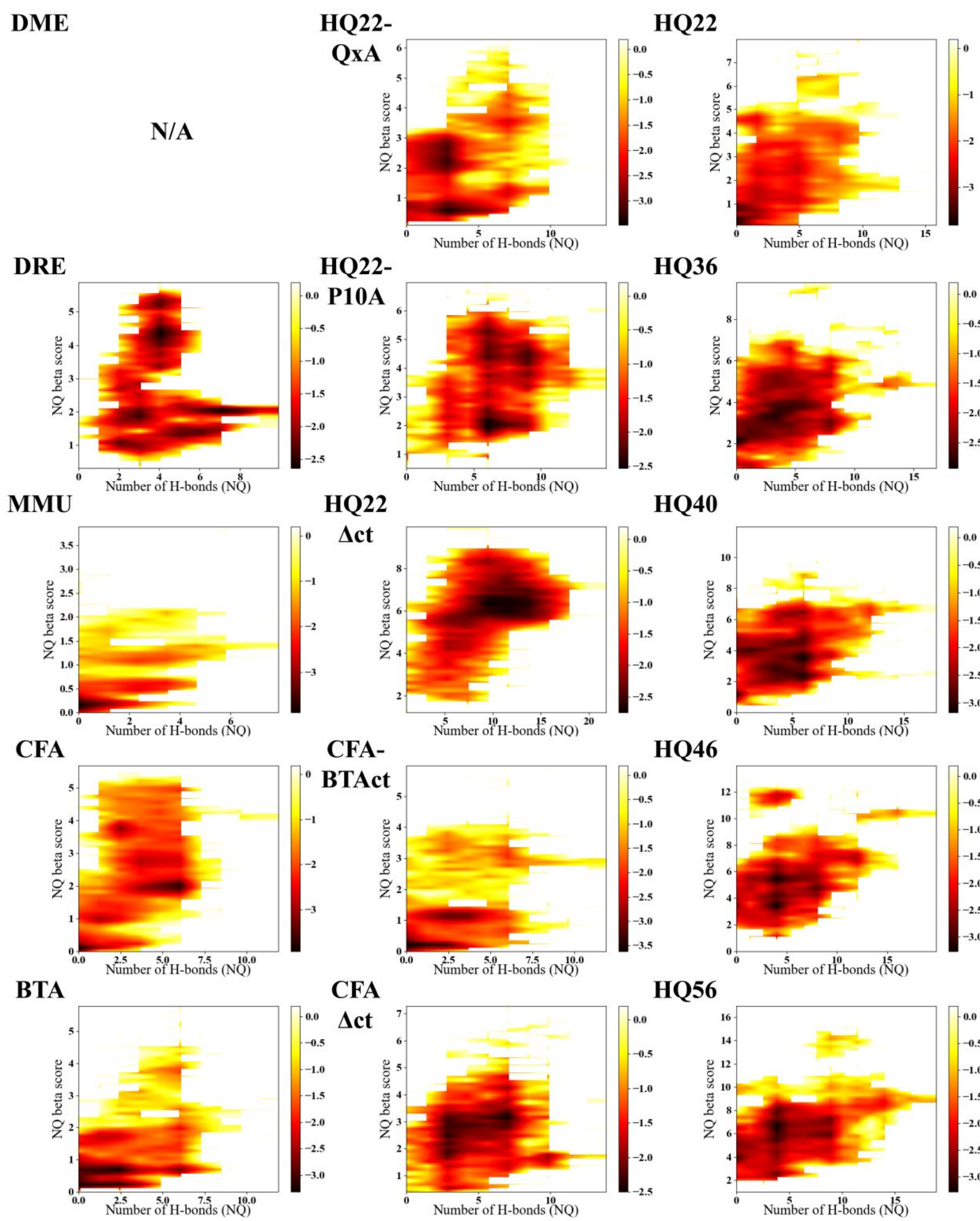

Figure S18. Free energy landscape calculated with a Boltzmann function using the NQ beta score and the number of NQ H-bonds as order parameters. DME was left out due to the absence of its polyQ.

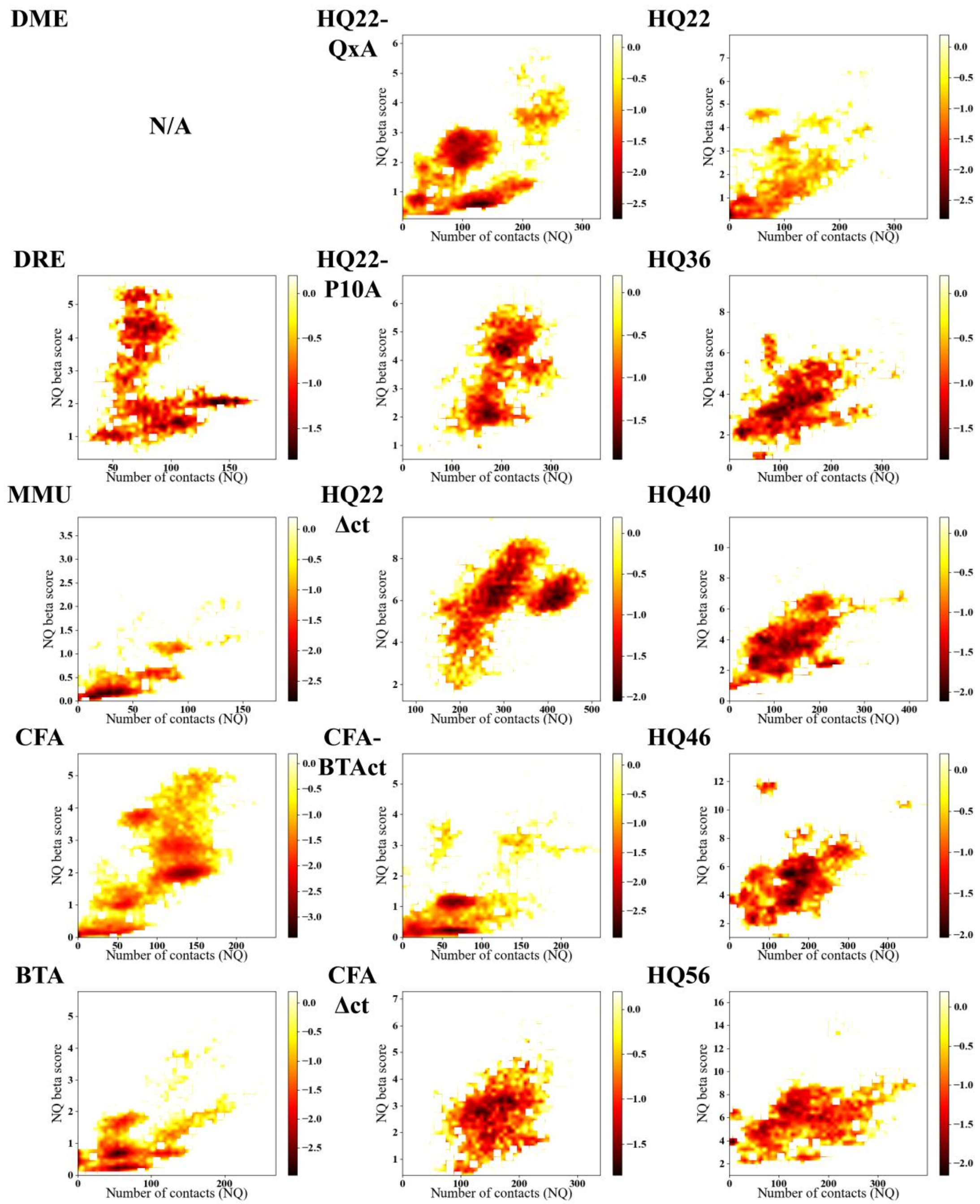

Figure S19. Free energy landscape calculated with a Boltzmann function using the NQ beta score and the number of NQ contacts (within 4.5 Å) as order parameters. DME was left out due to the absence of its polyQ.

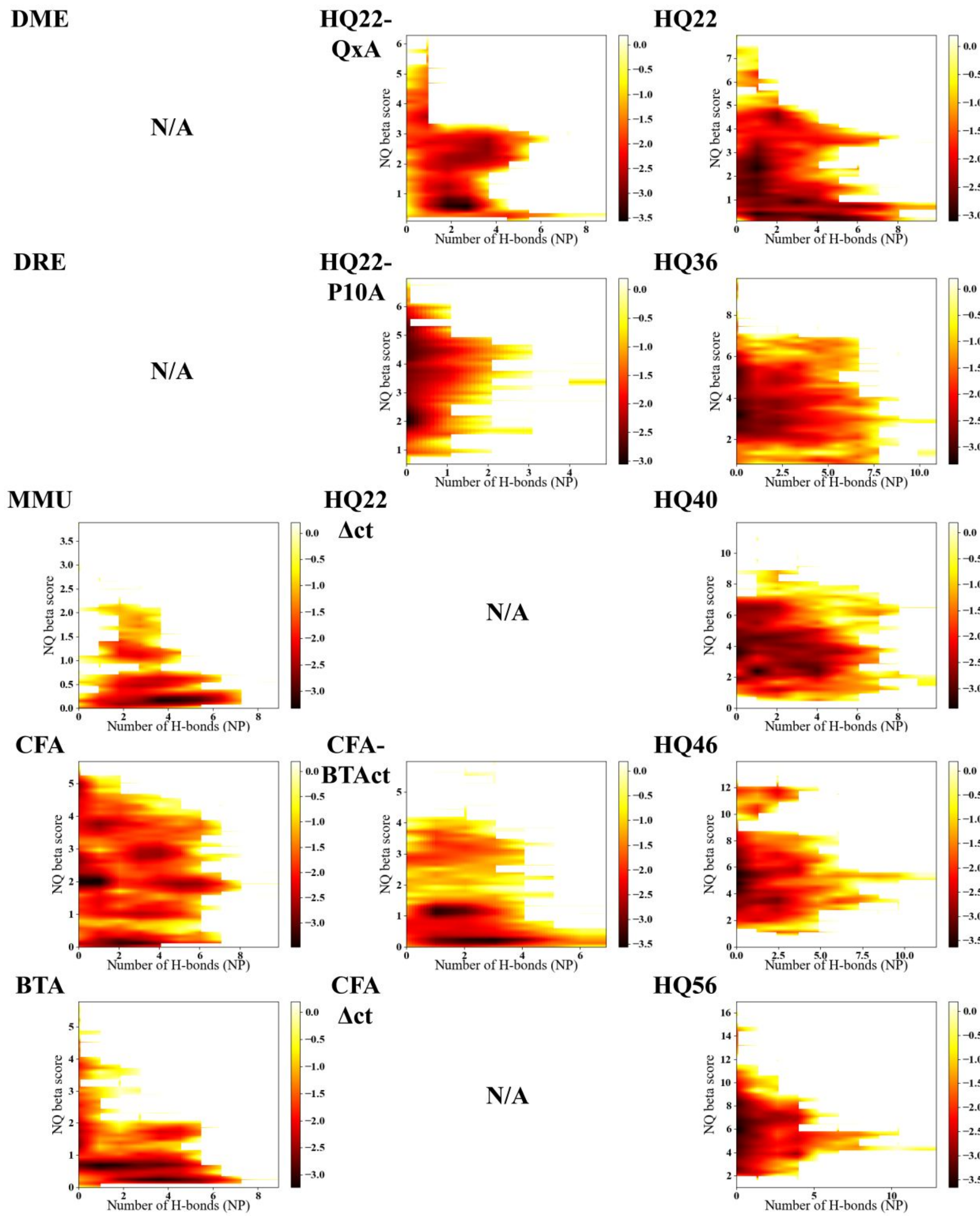

Figure S20. Free energy landscape calculated with a Boltzmann function using the NQ beta score and the number of NP H-bonds as order parameters. DME, DRE, HQ22Δct and CFAΔct were left out due to the absence of their PRDs.

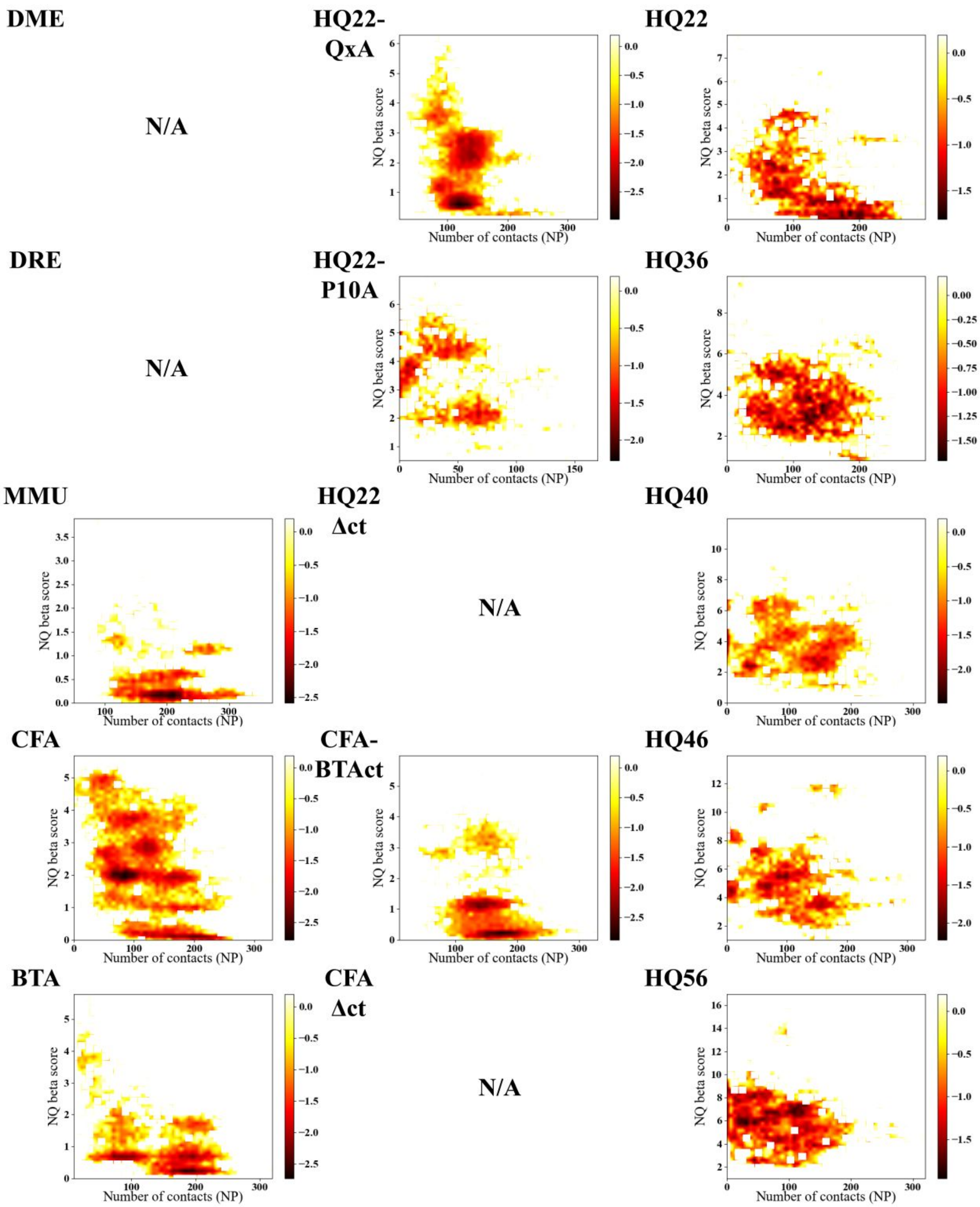

Figure S21. Free energy landscape calculated with a Boltzmann function using the NQ beta score and the number of NP contacts (within 4.5 Å) as order parameters. DME, DRE, HQ22Δct and CFAΔct were left out due to the absence of their PRDs.

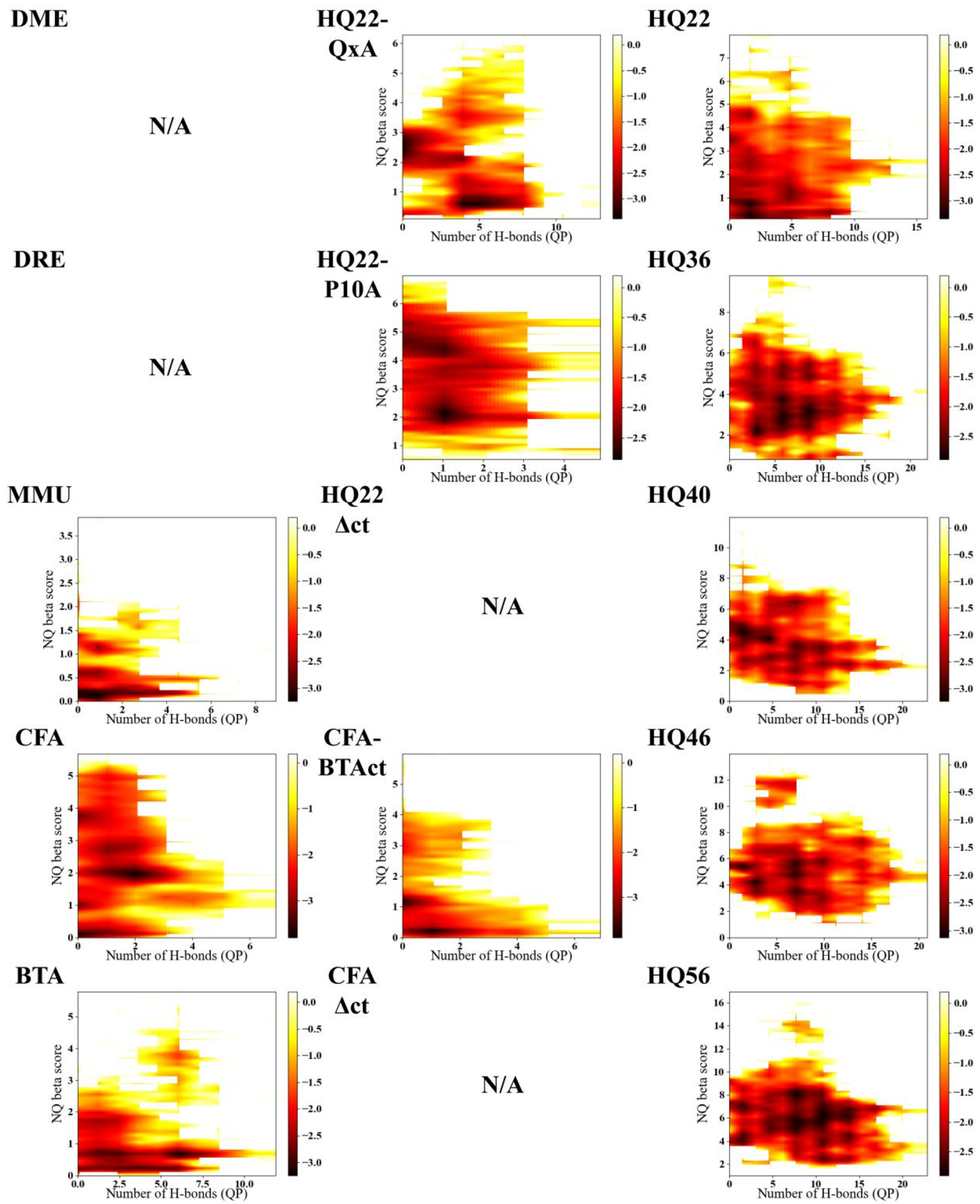

Figure S22. Free energy landscape calculated with a Boltzmann function using the NQ beta score and the number of QP H-bonds as order parameters. DME, DRE, HQ22Δct and CFAΔct were left out due to the absence of their PRDs.

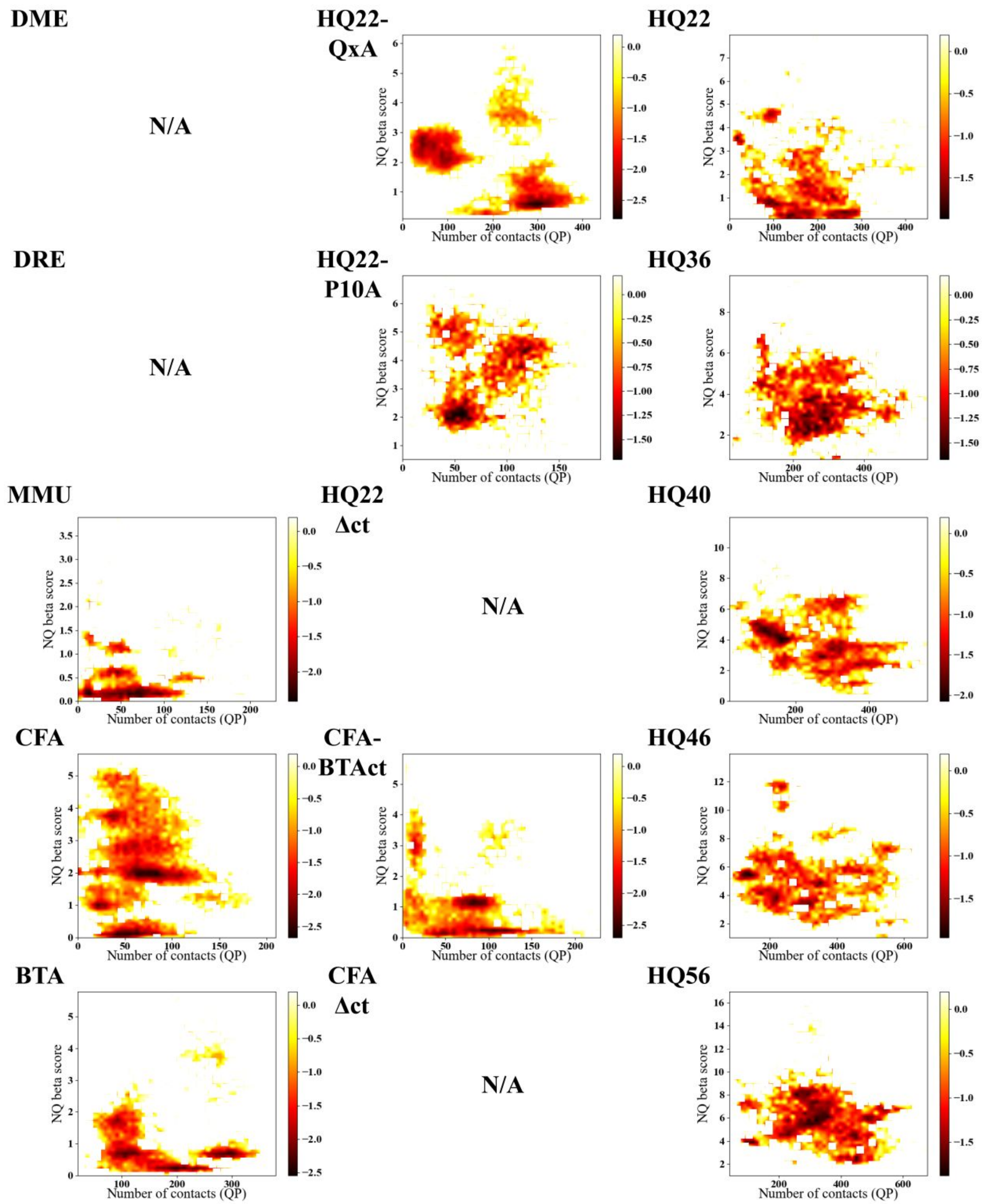

Figure S23. Free energy landscape calculated with a Boltzmann function using the NQ beta score and the number of QP contacts (within 4.5 Å) as order parameters. DME, DRE, HQ22Δct and CFAΔct were left out due to the absence of their PRDs.

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Dentatorubropallidoluysian atrophy (DRPLA)
SP|P54259|ATN1_HUMAN LPPASSSAPAPPMRFPYSSSSSSSAAASSSSSSSSSSASPFPASQALPSYPHSFPPPTSL 418
SP|O35126|ATN1_MOUSE LPPASSSAPGPPMRYPYSSSSSS-AAASSSSSSSS--ASQYPASQALPSYPHSFPPPTSM 417
...
SP|P54259|ATN1_HUMAN HHHHHQQQQQQQQQQQQQQQQQQQHHGNSGPPPPGAFPHPLEGGSSHHAHPYAMSPSLGS 538
SP|O35126|ATN1_MOUSE HHHH-QQQPQQQ-----------HHHGNSGPPPPGAYPHPLESSNSHHAHPYNMSPSLGS 524

Spinal and bulbar muscular atrophy (SBMA)
SP|P10275|ANDR_HUMAN GASLLLLQQQQQQQQQQQQQQQQQQQQQQQETSPR-QQQQQQ--GEDGSPQAHRRGPTGY 107
SP|O97775|ANDR_PANTR GASLLL-QQQQQQQQQQQQQQQQQQQQQQ-ETSPR-QQQQQ---GEDGSPQAHRRGPTGY 104
SP|Q9GKL7|ANDR_PIG   GARL---QQQQLQQQ---------------ETSPRRQQQQQQPSEDGSPQVQSRGPTGY 92
SP|Q9TT90|ANDR_CANLF GAHL---QQQQQQQQQQ-------------ETSPR-QQQQQQQ-GDDGSPQAQSRGPTGY 92
SP|P19091|ANDR_MOUSE GACL---QQRQ-------------------ETSPRRRRRQQHTE--DGSPQAHIRGPTGY 86
...
SP|P10275|ANDR_HUMAN QLYGPCGGGGGGGGGGGGGGGGGGGGGGGEAGAVAPYGYTRPPQGLAGQESDFTAPDVWY 504
SP|O97775|ANDR_PANTR QLYGPCGGGGGGGGGGGGGGGGG------EAGAVAPYGYTRPPQGLAGQEGDFTAPDVWY 495
SP|Q9GKL7|ANDR_PIG   QLYGPCGGGGGG--------------SAGEAGAVAPYGYTRPPQGLAGQEGDLAIPDIWY 480
SP|Q9TT90|ANDR_CANLF QLYGPCGGSGGG--------------SAGDGGSVAPYGYTRPPQGLAGQEGDFPPPDVWY 491
SP|P19091|ANDR_MOUSE QLYGP-GGGGG-------------SSSPSDAGPVAPYGYTRPPQGLTSQESDYSASEVWY 483

Spinocerebellar ataxia Type 7 (SCA7)
SP|O15265|ATX7_HUMAN MSERAADDVRGEPRRAAAAAGGAAAAAARQQQQQQQQQQPPPPQPQRQQHPPPPPRRTRP 60
SP|Q8R4I1|ATX7_MOUSE MSERAADDVRGEPRRAA---GGAAAA--RQQQQQ-----PQPLQPQRQ-HPP--LRRPRA 47
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Table S1. Sequence alignments for atrophin-1 (ATN1), androgen receptor (ANDR) and ataxin 7 (ATX7). Possible co-slippage (duplication of short nucleotide repeats at several loci simultaneously caused by replication slippage) events are highlighted in yellow.

References

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